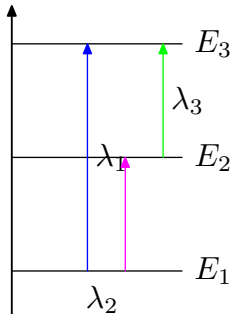


Absorption

Emission

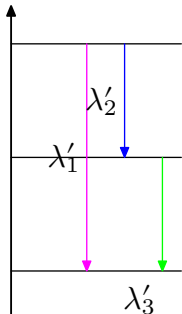
 E  E_3 E_2 E_1 λ_2 E_2 E_1

$$\Delta E = E_3 - E_1$$

$$E_{\text{Photon}} = h \cdot f = \frac{h \cdot c}{\lambda_1}$$

$$\Delta E = E_3 - E_1$$

$$E_{\text{Photon}} = h \cdot f = \frac{h \cdot c}{\lambda'_1}$$

 λ'_2 λ'_1 λ'_3